# Case Study - 1

(1) Brightmart is a medium sized online shopping company. Customers use its website and mobile application to purchase products.

The Company ~~Store~~ Stores ~~Source~~ :-

1. Customer name
2. Email addresses
3. Phone Numbers
4. Delievery addresses
5. Passwords
6. Order Information
7. Payment-related information

Brightmart has around 100 employees.

The company recently hired several new employees. To make things easier, the IT Department gave employees access to different internal system.

However, the company has the following security weakness.

- Some employees use simple passwords such as BrightMart123
- Many employees use the same passwords for multiple systems.
- Software updates are sometimes delayed.
- The company does not regularly conduct employee cybersecurity training.
- There is no multi-factor authentication for some important accounts.

- Employees can access sensitive information that they do not necessarily need for their jobs.
- Backups are performed only once a month.

## The Incident

One monday morning, an employee named Rahul receives an email.

Subject: Urgent - Your Account Will Be Suspended.
The email appears to come from the company's IT department.

It says:
"Your company account will be suspended today. Click the link below and verify your password immediately".

Rahul clicks the link.
The page looks almost identical to the company's login page.
He enters his username and password.
Rahul does not realize that the websit is controlled by an attacker.
The attacker now has Rahul's credendials.

## Stage 1 - Unauthorized Access.

The attacker uses Rahul's credentials to access BrightMart's internal system.

Because Rahul uses the same password for another system, the attacker is also able to access that system.

The attacker discovers that Rahul has access to customer information.

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The attacker downloads a large amount of customer data.

Q) What cybersecurity problem has occurred?

Stage 2 - Malware Attack

The attacker sends another email to Rahul containing a malicious attachment.

The malware begins communicating with the attacker's system.

The attacker now attempts to move from Rahul's computer to other systems within BrightMart's network.

Q) Which vulnerability allowed the attacker to move further into the organization?

Stage 3 - Ransomware

Two days later, employees arrive at work & discover that they cannot access important files.
A message appears on their computer:-
"Your Files have been encrypted. Pay the ransom to recover them".

The company's customer database and several important business files are encrypted.
Unfortunately, the company's most recent backup is almost a month old.
BrightMart's website stops working.
Customers cannot place orders
The customer-support team cannot access customer records.

Stage 4 - Business Impact
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BrightMart now faces several problems.
Financial Impact
The company loses sales because its website is unavailable
Data Impact
Customer information may have been stolen.
Operational Impact
Employees cannot access important systems.
Reputation Impact
Customers lose confidence in the company.
Legal / Compliance Impact
The company may face regulatory or legal consequences depending on what information was exposed & which laws apply.

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Q) What cybersecurity problem has occured?
→ Possible answers:-
- Phishing
- Credential theft
- Unauthorized access.
- Data breach

Q) Which vulnerability allowed the attacker to move further into the organization?
→ Students should identify possibilities such as:
- Weak passwords
- Excessive user privileges
- Lack of MFA
- Poor network security
- Lack of employee awareness
- Outdated software.

Q) What threats are present in this case?
→ They may identify:
- Phishing
- Malware
- Ransomware
- Credential theft
- Unauthorized access.

Data theft.

Q) Who is responsible for security?
→ Not only the IT Dept. Every employee who uses the organizations systems has a role in cybersecurity.

Q) What weakness in BrightMart's security allowed the attack to succeed?

→

| Vulnerability | How it helped the attacker |
|---|---|
| Weak passwords | Easier to compromise accounts |
| Password reuse | One stolen password gives access to multiple systems. |
| NO MFA | Stolen password may be enough to log in |
| ~~Lat~~ Lack of training | Employee falls for phishing |
| Delayed updates | Vulnerable software may be explaited. |
| Excessive privileges | Attacker can access more data. |

Q) Was confidential info. exposed?

→ Yes. Customer information was accessed/downloaded by the attacker.

Q) Was info. changed or damaged?

→ Yes. The attacker could modify or encrypt files.

Q) Were systems available to legitimate users?

→ NO. The website & internal systems became unavailable because of ransomware.

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Group 4 –

Q) What should BrightMart do to prevent this from happening again?

→
- Use strong passwords
- Provide regular cybersecurity awareness training.
- Keep software updated
- Segment imp. networks
- Encrypt sensitive data
- Deploy antivirus / endpoint protection.
- Monitor suspicious login activity.
- Develop an incident response plan.

Q) was phishing the only problem?

→ No, Phishing was the initial attack, but weakeness such as password reuse, lack of MFA, excessive privileges, poor backups.

Q) What was the biggest vulnerability?

→ There is no single correct answer.
Students should justify their choice.
For ex.
"Lack of employee awareness was a major vulnerability because the attack started when Rahul clicked the phishing link".
Another student might argue:
"Poor backup practices were the biggest problem because BrightMart could not quickly recover from ransomware."
Both can be valid if properly justified.

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